

ative MRR added this month



SQLMentor: User Submission Analysis

A comprehensive SQL mini project designed to analyze user submission data from a competitive programming platform, extracting meaningful insights through structured queries and real-world analytics.

Core Objectives



Real-World SQL Practice

Practice querying structured data using industry-standard SQL techniques and best practices.



Performance Analysis

Analyze user performance based on submissions, points earned, and scoring trends over time.



Time-Based Insights

Work with temporal data to generate daily and weekly analytical reports.



Leaderboard Analytics

Generate competitive leaderboard insights using advanced SQL ranking functions.

This project strengthens essential SQL concepts including GROUP BY, ORDER BY, COUNT, SUM, AVG, and window functions—perfect for beginners to intermediate learners seeking hands-on experience.

Dataset Overview

Source File

user_sub_sql_mentor06nov.csv

Contains comprehensive records of user submissions on the SQLMentor platform. Each row represents a single submission made by a user for a particular question.

Database Table

user_submissions

Stores all submission records and serves as the foundation for analytical queries.

Column Structure

- **id** – Unique identifier for each submission
- **user_id** – Unique ID of the user
- **username** – Name of the user
- **question_id** – ID of the question attempted
- **points** – Points earned for the submission
- **submitted_at** – Date and time of submission

Book

isbn	varchar(50)
title	varchar(100)
pages	integer
price	money
description	varchar(256)
publisher	varchar(100)

Book_details

id	integer
book_isbn	varchar(50)
rating	decimal
language	varchar(10)
keywords	text
date_publisher	date

QUERY #1

User Statistics Overview

Objective

List all distinct users with comprehensive performance statistics to understand overall platform engagement and achievement levels.

Output Metrics

- Display each user's name
- Total number of submissions
- Total points earned

Purpose: Get an overall performance summary of every user on the platform for holistic analysis.



QUERY #2

Daily Average Points Analysis

OUR HR PERFORMANCE METRICS CALENDAR TEMPLATE IS NOT JUST A CALENDAR; IT'S A COMPASS GUIDING YOU TOWARDS STRATEGIC SUCCESS, ONE METRIC AT A TIME.

April [1]		2024		HR PERFORMANCE METRICS CALENDAR						
April										
SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY				
31	1	2	3	4	5	6				
7	8	9	10	11	12	13				
14	15	16	17	18	19	20				
21	22	23	24	25	26	27				
28	29	30	1	2	3	4				
5	6	7	8	9	10	11				
Employee Name	Date	Department	Metric Evaluation	Training Hours	Comments	Recommendations				
David Turner	10 Apr 2024	Customer Service	Attendance: 100% Analysis	5	Addressed customer complaints with empathy and efficiency.					

Daily Performance Tracking

Compute the average points scored by each user per day to analyze performance consistency and identify scoring trends over time.

Consistency Insights

This query reveals which users maintain steady performance versus those with fluctuating daily scores, helping identify learning patterns.

Daily Top Performers Leaderboard



Daily Ranking System

Rank users daily based on positive (successful) submissions and display only the top 3 users per day.

Key Features

- Identifies daily champions
- Tracks successful submission counts
- Creates competitive leaderboard
- Motivates consistent performance

Purpose: Create a daily leaderboard to identify top performers and encourage healthy competition among users.

Identifying Struggle Areas

1

Query Objective

Identify users with the most incorrect or zero-point submissions and display the top 5 users who need additional support.

2

Analytical Value

Analyze areas where users struggle most frequently to understand common pain points and knowledge gaps.

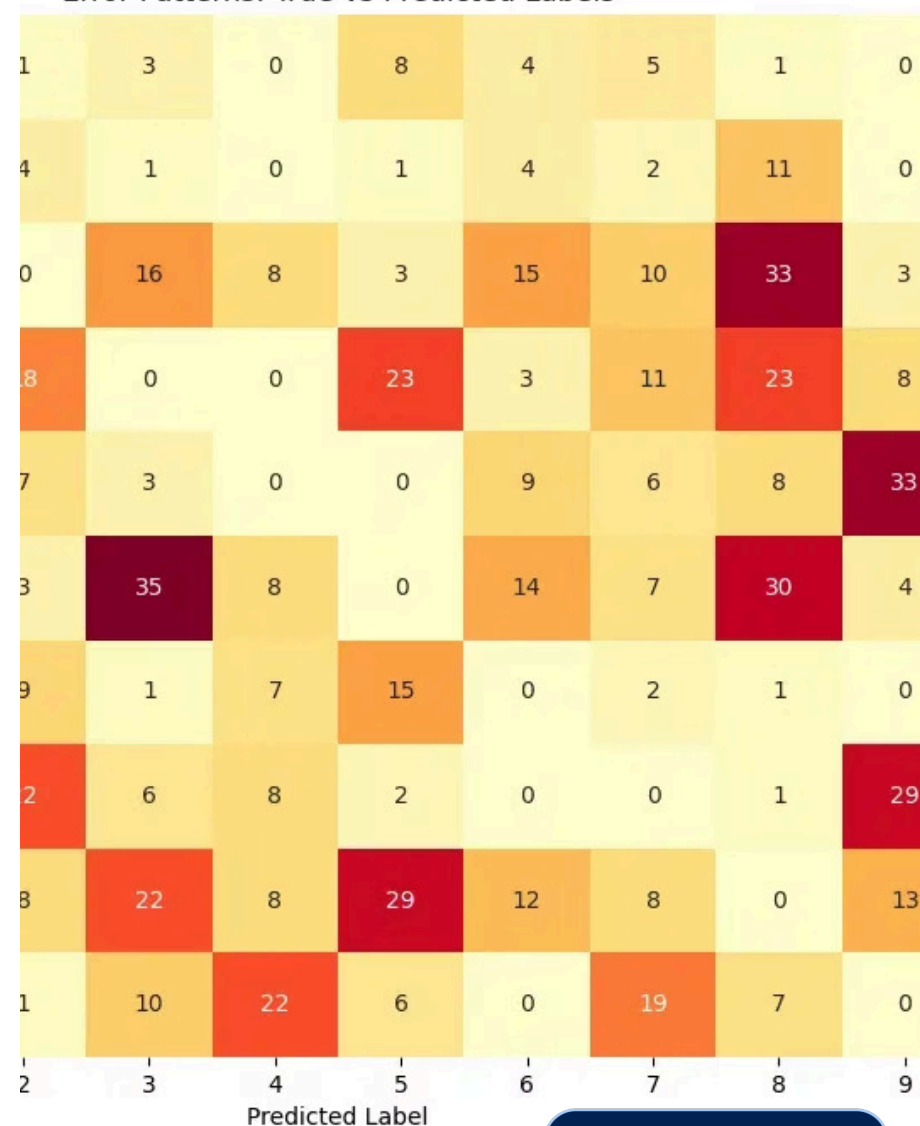
3

Actionable Insights

Identify improvement needs and opportunities for targeted educational interventions or additional learning resources.

Impact: This analysis helps platform administrators provide personalized support and improve learning outcomes for struggling users.

Error Patterns: True vs Predicted Labels



Weekly Performance Rankings

Weekly Aggregation

Aggregate user points on a weekly basis, rank users by total weekly performance, and display the top 10 performers for each week.

Long-Term Evaluation

This query evaluates sustained performance over longer periods, revealing which users demonstrate consistent excellence versus sporadic success.

ANALYZING LEAGUE TABLE RANKINGS: A CLOSER LOOK AT THE TOP PERFORMERS

Pos	Team	Pld	W	D	L	GF	GA	GD	Pts
1	Manchester City	38	32	4	2	85	20	65	100
2	Liverpool	38	29	7	2	89	22	67	94
3	Chelsea	38	22	9	7	63	28	35	75
4	Arsenal	38	21	7	10	74	49	25	70
5	Tottenham	38	20	9	9	66	36	30	69
6	Manchester United	38	18	9	11	60	44	16	63
7	Leicester City	38	15	7	16	51	48	3	52
8	Everton	38	14	7	17	50	46	4	49
9	West Ham	38	13	10	15	47	53	-6	49
10	Leeds United	38	14	5	19	50	56	-6	47

Purpose: Evaluate long-term performance patterns and weekly consistency to identify truly dedicated and skilled users on the platform.

SQL Concepts Mastered



Core SQL Operations

SELECT, WHERE, GROUP BY, ORDER BY, and HAVING clauses for data filtering and organization.



Aggregate Functions

COUNT(), SUM(), AVG() for statistical analysis and data summarization.



Date & Time Functions

Temporal data manipulation for daily and weekly analysis.



Advanced Queries

Subqueries and Common Table Expressions (CTEs) for complex analysis.



Window Functions

RANK() and DENSE_RANK() for leaderboard and ranking systems.

DATA ANALYSIS

A COMPREHENSIVE



Project Impact & Career Readiness

5

Complex Queries

Analytical problems solved using real-world data scenarios

8+

SQL Concepts

Core and advanced techniques mastered through hands-on practice

100%

Practical Skills

Industry-relevant experience in data analysis and reporting

The SQLMentor project provides comprehensive hands-on exposure to real-world SQL analytics problems. By working with user submission data, this project demonstrates how SQL can be effectively used to extract insights, evaluate performance, and support data-driven decision-making.

"This project serves as a strong foundation for roles such as Data Analyst, Business Analyst, and SQL Developer—building the essential skills employers seek in data professionals."